

Assignment 8: GLMs (Linear Regressios, ANOVA, & t-tests)

Lex Clay

Spring 2026

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

NOTE: For this assignment, you can skip tests for any of the assumptions required to run the models.

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1

#install.packages("agricolae")

library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.6.0
## v ggplot2    4.0.0      v tibble    3.3.1
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
library(here)
```

```
## here() starts at C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
#2
```

```
NTL.LTR.Raw <- read.csv(here("./Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"))
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: Depth across all lakes has no effect on lake temperature recorded in July. Ha: Lake temperature recorded in July decreases with depth across all lakes.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#Extract Date to get July
```

```
NTL.LTR.Raw$sampledate = ymd(NTL.LTR.Raw$sampledate)
```

```
## Warning: 33138 failed to parse.
```

```
NTL.LTR.Raw$month <- month(NTL.LTR.Raw$sampledate)
```

```
#4
```

```
#wrangling the raw data set and then applying lm()
```

```
NTL.LTR.wrangled <- NTL.LTR.Raw %>%
```

```

select(lakename, year4, daynum, depth, temperature_C, month) %>%
filter(month==07) %>%
na.omit()

#5

# Noodling with this version that adds a linear regression line AFTER the plot is created but not to th

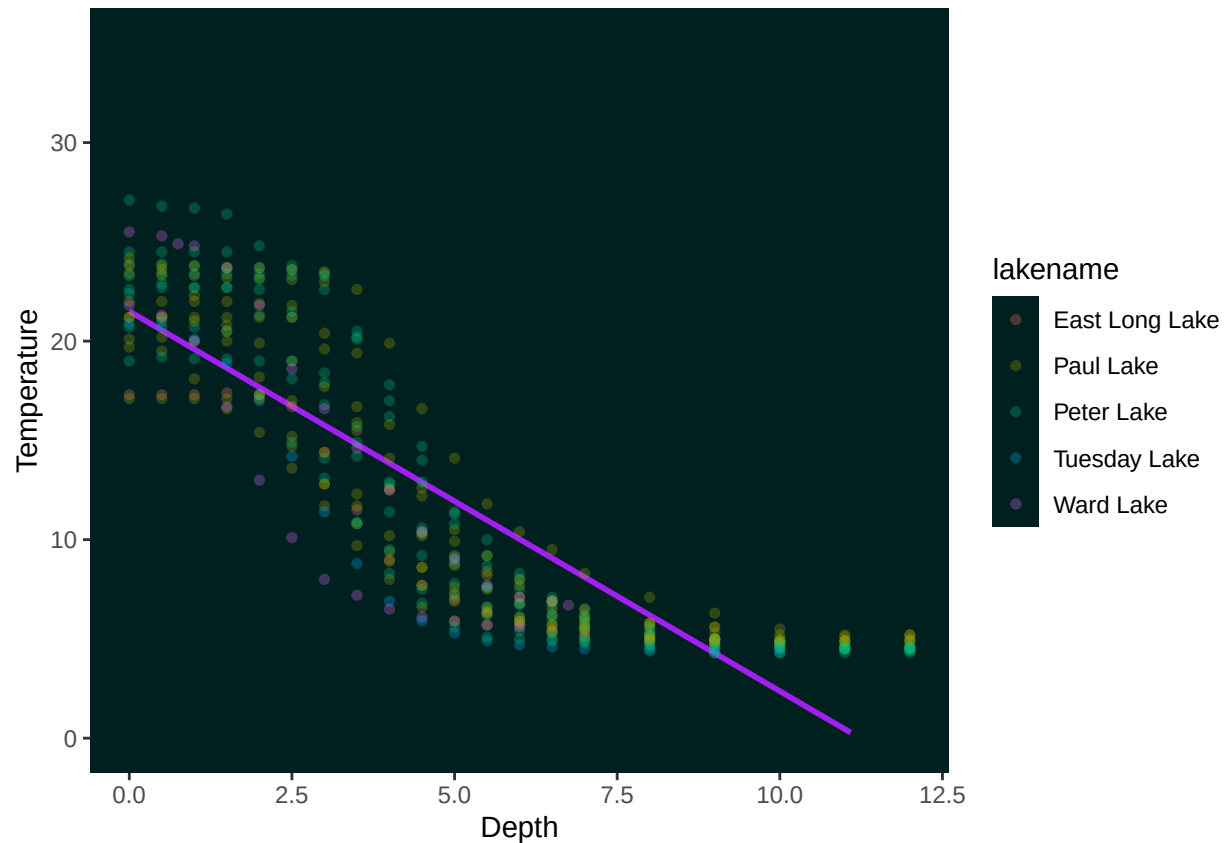
July.NTL.LTR.scatter <- ggplot(NTL.LTR.wrangled, aes(x=depth, y=temperature_C))+labs(
  x="Depth",
  y="Temperature"
)+
  ylim(0,35)+
  theme(
    panel.background = element_rect(fill = "#002020"),
    panel.grid = element_blank()
  )+
  geom_smooth(method="lm", se=FALSE, color="purple")+
  geom_point(aes(color=lakename), alpha=0.3, size=1.3)

print(July.NTL.LTR.scatter)

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_smooth()').

```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The figure(s) suggest(s) that there is a inverse relationship between depth and temperature. As depth increases, temperature tends to decrease. The intercept expects that when the value of the depth is 0 - so at the surface - the temperature will be 21.5 degrees celsius.

7. Perform a linear regression to test the relationship and display the results.

```
#7
July.Reggression.NTL.LTR <- lm(data=NTL.LTR.wrangled, temperature_C ~ depth)
summary(July.Reggression.NTL.LTR)

##
## Call:
## lm(formula = temperature_C ~ depth, data = NTL.LTR.wrangled)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.7641 -2.8586 -0.3779  2.6155  7.7928
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.5054      0.3261   65.95  <2e-16 ***
## depth       -1.9138      0.0567  -33.76  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.65 on 392 degrees of freedom
## Multiple R-squared:  0.744, Adjusted R-squared:  0.7434
## F-statistic: 1139 on 1 and 392 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The slope indicates that with each depth increase of 1, the temperature decreases by -1.9. The P value is 2.2e-16, which is basically 0 == statistically significant.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9

#the smaller the AIC value, the better the model performance is. AIC is run by default
#using the function step(). The best model uses as few predictors as possible.

#running the linear regression
temp.predictors <- lm(data=NTL.LTR.wrangled, temperature_C~year4+daynum+depth)

#run AIC
my.aic <-step(temp.predictors)

## Start:  AIC=966.18
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS    AIC
## - year4    1      21.7 4505.8  966.08
## <none>                 4484.0  966.18
## - daynum    1     638.3 5122.3 1016.62
## - depth     1    15263.4 19747.5 1548.29
##
## Step:  AIC=966.08
```

```
## temperature_C ~ daynum + depth
##
##           Df Sum of Sq      RSS      AIC
## <none>                4505.8   966.08
## - daynum    1         717   5222.8 1022.27
## - depth     1       15242 19747.5 1546.29
```

#10

#running the multiple linear regression w/ depth and daynum because AIC sky rockets when either of those are removed

```
good.model <- lm(data=NTL.LTR.wrangled, temperature_C~daynum+depth)
summary(good.model)
```

```
##
## Call:
## lm(formula = temperature_C ~ daynum + depth, data = NTL.LTR.wrangled)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.6397 -2.6632 -0.3101  2.5174  8.0771
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 11.198599   1.341363   8.349 1.21e-15 ***
## daynum       0.054660   0.006929   7.888 3.10e-14 ***
## depth       -1.917670   0.052729  -36.368 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.395 on 391 degrees of freedom
## Multiple R-squared:  0.7792, Adjusted R-squared:  0.778
## F-statistic: 689.8 on 2 and 391 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

> Answer: I ran the the linear regression above w/ depth and daynum because AIC sky rockets when either of those are removed. Meanwhile year4 AIC is the same as so it has no stat. influence. This model - good.model - has a higher R-squared value than the model using only depth (.778 vs .743) and therefore is a better indicator of temperature variance.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

#Wrangle

```
lake.temps <- NTL.LTR.wrangled %>%
```

```
  group_by(lakename) %>%
```

```
  summarise(average_temp=mean(temperature_C))
```

```
summary(lake.temps) #this shows the average temps for each lake but now we need to use anova/lm to dete
```

```
##   lakename      average_temp
```

```
## Length:5      Min.   :10.12
```

```
## Class :character 1st Qu.:10.56
```

```
## Mode  :character Median :12.46
```

```
##              Mean   :11.92
```

```
##              3rd Qu.:12.56
```

```
##              Max.   :13.93
```

```
anova.temp <- aov(data=NTL.LTR.wrangled, temperature_C ~ lakename)
```

```
lm.model <- lm(data=NTL.LTR.wrangled, temperature_C ~ lakename)
```

```
summary(anova.temp)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
```

```
## lakename      4      228    56.96   1.098  0.357
```

```
## Residuals    389    20176    51.87
```

```
summary(lm.model)
```

```
##
```

```
## Call:
```

```
## lm(formula = temperature_C ~ lakename, data = NTL.LTR.wrangled)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -8.331 -6.756 -2.550  7.338 14.536
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)    10.5556     1.6975   6.218 1.3e-09 ***
```

```
## lakenamePaul Lake     1.9008     1.7856   1.065  0.288
```

```
## lakenamePeter Lake    2.0088     1.7904   1.122  0.263
```

```
## lakenameTuesday Lake -0.4389     2.4006  -0.183  0.855
```

```
## lakenameWard Lake     3.3755     2.1610   1.562  0.119
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 7.202 on 389 degrees of freedom
```

```
## Multiple R-squared:  0.01117,    Adjusted R-squared:  0.0009981
```

```
## F-statistic: 1.098 on 4 and 389 DF,  p-value: 0.3571
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: A low F-value of 1.098 indicates that the difference in mean temperature among the lakes varies about as much as the temperature within the lakes and the the P value .357 suggests that temperature variance between lakes is likely random. The R-squared value indicates that the lake name only accounts for about 1% of temperature reading. Ergo, not statistically significant.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

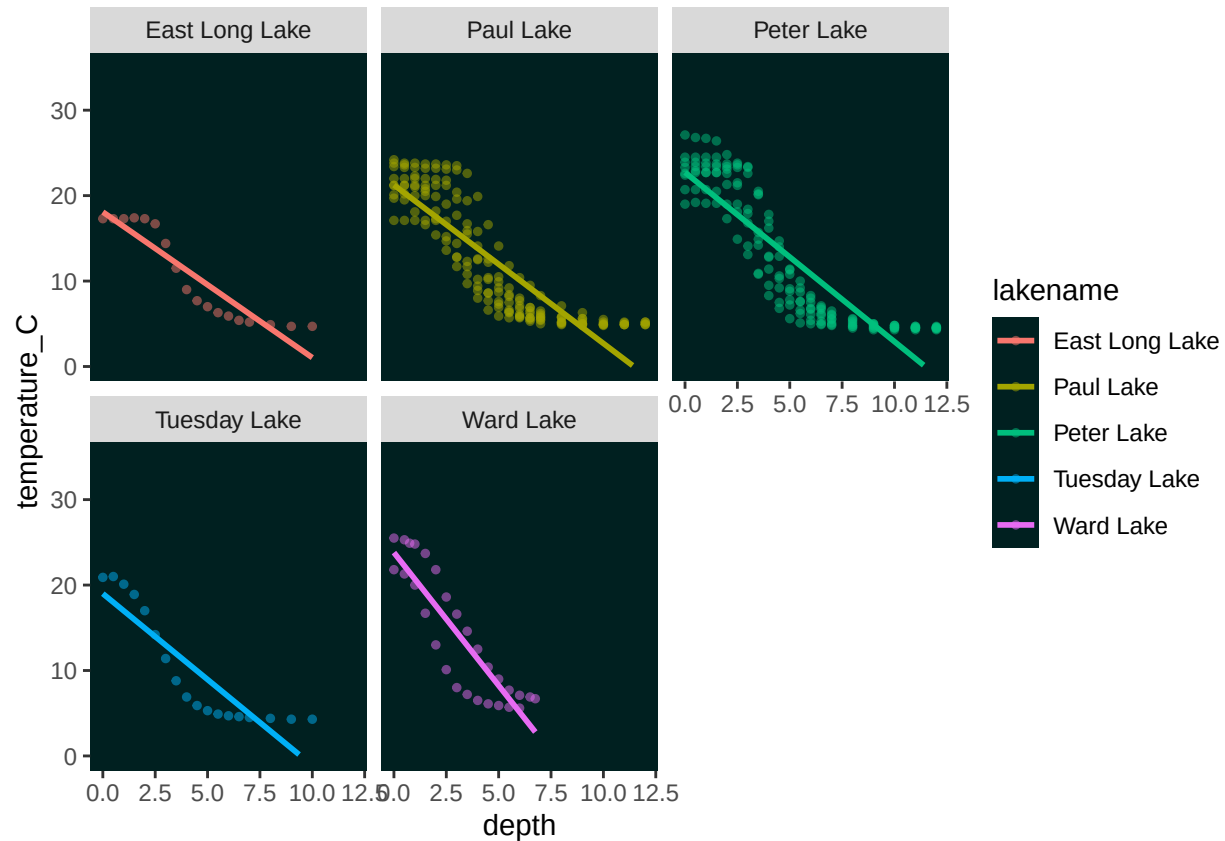
```
#14

NTL.LTR.faceted <- ggplot(NTL.LTR.wrangled, aes(x=depth, y=temperature_C, color=lakename))+
  geom_point(alpha=0.5, size=1)+
  geom_smooth(method="lm", se=FALSE)+
  ylim(0,35)+
  facet_wrap(~lakename)+
  theme(
    panel.background = element_rect(fill = "#002020"),
    panel.grid = element_blank()
  )

#still need to remove panel grid here
print(NTL.LTR.faceted)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 13 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
TukeyHSD(anova.temp)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = NTL.LTR.wrangled)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Paul Lake-East Long Lake	1.9007758	-2.992848	6.794400	0.8245987
Peter Lake-East Long Lake	2.0088194	-2.898035	6.915674	0.7948864
Tuesday Lake-East Long Lake	-0.4388889	-7.018014	6.140236	0.9997496
Ward Lake-East Long Lake	3.3754789	-2.546994	9.297952	0.5227817
Peter Lake-Paul Lake	0.1080436	-2.069085	2.285172	0.9999228
Tuesday Lake-Paul Lake	-2.3396647	-7.233289	2.553959	0.6849800
Ward Lake-Paul Lake	1.4747031	-2.492456	5.441862	0.8466692
Tuesday Lake-Peter Lake	-2.4477083	-7.354562	2.459146	0.6491273
Ward Lake-Peter Lake	1.3666595	-2.616808	5.350127	0.8810112
Ward Lake-Tuesday Lake	3.8143678	-2.108105	9.736840	0.3955788

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Since all lakes have a P value far above 0.05, none have a statistically distinct mean temperature. The lakes are pretty much indistinguishable via their temperature data.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: A T-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
#Crampton does not have any July data which is why it was dropped by na.omit() from the raw dataset. I'
```

```
ward.tuesday <- NTL.LTR.wrangled %>%  
  filter(lakename %in% c("Ward Lake", "Tuesday Lake"))  
  
ward.tuesday.test <- t.test(data=ward.tuesday, temperature_C ~lakename)  
  
ward.tuesday.test
```

```
##  
## Welch Two Sample t-test  
##  
## data: temperature_C by lakename  
## t = -1.8432, df = 38.793, p-value = 0.07295  
## alternative hypothesis: true difference in means between group Tuesday Lake and group Ward Lake is not equal to 0  
## 95 percent confidence interval:  
## -8.0008238 0.3720881  
## sample estimates:  
## mean in group Tuesday Lake mean in group Ward Lake  
## 10.11667 13.93103
```

```
#mean in group Tuesday Lake mean in group Ward Lake  
# 10.11667 13.93103
```

```
# Ward Lake-Tuesday Lake 3.8143678
```

Answer: The Tukey test says the diff between Ward Lake-Tuesday Lake is 3.814, which matches the t.test results to the T. (Sorry, I had to.)